Condensed Focus Group Transcript

Editorial note.

> This transcript has been edited for inclusion in the replication package. Repetitions, conversational coordination, and side exchanges were removed. The transcript is intended as a content-preserving condensed transcript, rather than a verbatim record.
>
> Speaker identifiers were retained to preserve traceability. **ORG_00** and **ORG_01** acted as organizers/mediators and should therefore be distinguished from participant contributions during qualitative analysis.
>
> 📝*indicates organizers' notes.*
>
> **Insight: …** *indicates summarized insights obtained from the discussion*

Participant and Speaker Mapping

Speaker ID	Role in Focus Group	Background	Experience (years)
ORG_00	Organizer / Mediator	Software Engineer (Research)	
ORG_01	Organizer / Mediator	Software Engineer (Research)	
P_00[AI-Ethicist]	Participant	AI Ethicist (Industry)	
P_01[Lawyer]	Participant	Lawyer (Research)	
P_02[Anthropologist]	Participant	Anthropologist (Research)	
P_03[SoftwareEng]	Participant	Software Engineer (Research)	
P_04[DigitalHumanity]	Participant	Digital Humanities (Research)	
P_05[Developer]	Participant	Developer (Industry)	

Study objective and scope

📝 The objectives of the study and the ones of the focus group were explained to participants.

ORG_00: The objective is to investigate whether an LLM generating code can write it in a way that reflects how a human developer would write it. Developers make decisions that are not completely determined by functional requirements, and some of these decisions can have ethical implications.

The study has three steps:

1. identify and validate relevant *soft ethics* dimensions;
2. determine how to elicit a developer's ethical profile;
3. understand how this profile should be represented in the prompt used for code generation.

The focus is specifically on the space left open by functional requirements, where the developer has discretion over implementation decisions.

---

1. Ethical dimensions

📝Participants were shown the table reporting the ethical concerns in its initial formulation.

Q1: Do these aspects represent a meaningful ethical consideration in software development?

📝All participants agreed.

Comments:

P_02[Anthropologist]: A valuable aspect of the descriptions and rationales is that they emphasize the role of the developer as someone who *selects*, *prioritizes*, or *avoids* particular alternatives. The developer's choice should remain explicit in the conceptualization.

ORG_00: The rationale behind this choice is that developers' choices are not necessarily conscious or explicit. Developers may make them implicitly as part of ordinary implementation decisions.

P_02[Anthropologist]: Even when the reasoning is not explicit, a choice is still being made. Intentionality may vary in degree, but the resulting implementation still reflects a selection among alternatives.

P_05[Develpoper]: An unconscious choice is always made. One does not think about it and was not taught to do it, it's something that arises unconsciously.

Insight: Developer choice is central in the definitions

Q2: Are the definitions sufficiently clear?

📝Participants agreed, but some comments were raised

Comments:

P_00[AI-Ethicist] / P_04[DigitalHumanity] / P_02[Anthropologist]: The definitions are generally clear and distinguishable.

P_02[Anthropologist]: Examples should be included at the end of each rationale, as already done for Transparency and Explainability. Concrete examples make the intended meaning easier to understand.

Insight: Examples are a key element for clarity

P_00[AI-Ethicist]: Fairness is currently expressed only in terms of avoiding undesirable outcomes. It should not only be about *avoiding unfairness*, but also about actively *promoting fairness*.

ORG_00: Therefore, the formulation should not be exclusively negative ("avoid something unfair") but should also capture the positive pursuit of fairness.

P_00[AI-Ethicist] / P_02[Anthropologist]: Agreed.

Insight: Fairness should also be expressed positively

P_01[Lawyer]: The formulation should not mention only sensitive data. Personal data in general should be considered, with particular attention to sensitive data as a specially protected subset. Restricting the formulation to sensitive data could make the scope unnecessarily narrow.

Insight: Fairness should not be restricted to sensitive data

📝A terminology issue was raised for sustainability:

P_03[SoftwareEng]: *Resource consumption* may be preferable to *energy consumption*, because the concern is not necessarily limited to energy.

Q3: Are the aspects clearly distinct from the others?

📝 Participants agreed, no further comments were raised.

Q4: Are there relevant aspects that are missing?

P_03[SoftwareEng]: Privacy appears to be missing as an independent ethical aspect.

ORG_00: One possibility is to remove *unjustified sensitive data collection* from fairness and introduce privacy as a separate category.

P_03[SoftwareEng]: Agreed.

P_01[Lawyer]: Privacy is a broad area, so the exact scope of this dimension should be made explicit.

Insight: Privacy as a separate dimension

Further discussion points

P_04[DigitalHumanity]: A developer's value may consist in preserving the values and agency of the final user. For example, when implementing a cookie-consent interface, a developer might value making the consequences understandable and making opt-out straightforward, including for less experienced users.
This suggests that a developer's ethical orientation may concern not only their own values but also their willingness to **preserve or enable the values and choices of end users**.
This aspect may overlap with inclusiveness, privacy, and transparency and should be clarified in the taxonomy.

P_01[Lawyer]: Inclusiveness should also be considered in relation to design decisions, rather than exclusively in terms of whether the final system "can be used" by different people.

Insight: preserving end-user values is a concern

📝A central issue in the discussion was distinguishing choices genuinely available to the developer from decisions imposed by requirements, management, design, legislation, or other stakeholders.

P_00[AI-Ethicist]: At what point in the system lifecycle is the framework intended to operate?

ORG_00: Only at implementation time: the developer already has the requirements and is writing the code. Decisions that are themselves functional requirements are outside the intended scope.

P_00[AI-Ethicist]: Therefore, the framework concerns the developer's remaining space of action.

P_03[SoftwareEng]: Many apparently ethical decisions may nevertheless not be under the developer's control. A designer or manager may have already determined an interaction pattern or technology.

P_02[Anthropologist]: The framework should explicitly define the **boundaries within which the developer can act**. It is not sufficient to say that the framework operates at implementation time. It should also clarify the assumptions and decisions already fixed upstream.
These boundaries form a perimeter around the developer's autonomy. The developer should only be considered responsible for decisions that actually fall within this perimeter.

ORG_00: This is particularly important for the final LLM-based system. If the entire existing codebase were used to infer the developer profile, the inferred behavior could reflect company-level decisions and constraints rather than the developer's own choices. The goal is instead to capture what the developer would choose within their actual discretionary space.

P_02[Anthropologist]: The perimeter should therefore be transparent. Otherwise it becomes difficult to distinguish choices attributable to the developer from choices attributable to external requirements.

Insight: The perimeter of the developer's choice should be transparent

📝**Implication**

The study should explicitly represent:

- decisions fixed by functional, organizational, legal, or technical requirements;
- the residual space in which the developer has autonomy;
- ethical preferences that operate only within that residual space.

The LLM should reproduce the developer's choices **only within this discretionary space**, rather than reproducing company-level or externally imposed decisions as though they represented the developer's ethical profile.

---

2. Questionnaire and profile elicitation

📝 The proposed questionnaire used multiple statements for each ethical dimension, rated on a five-point Likert scale. Participants were shown the statements.

📝 Discussion was triggered by participants before specific questions were asked. The following reports the discussed points.

P_00[AI-Ethicist]: For fairness and inclusiveness in particular, several statements appear difficult to discriminate. A respondent who agrees with one is likely to agree with the others, producing artificially coherent profiles.

P_03[SoftwareEng]: The same problem appears for fairness. Different formulations may nevertheless elicit essentially the same response.

Insight: Statements may be too similar

P_00[AI-Ethicist]: The statements should be made more specific by using examples of **developer actionability**: concrete situations in which developers can genuinely make different implementation choices.

P_05[Developer]: The technical examples are easier to understand from a developer's perspective. Abstract fairness or inclusiveness statements may describe things developers do not explicitly ask themselves while coding, whereas concrete implementation situations are more realistic.

P_02[Anthropologist]: One option is to foreground the concrete example or scenario and move the abstract statement to a secondary position, possibly in parentheses.
This would allow the developer to immediately recognize a realistic situation encountered during development.
However, explicitly associating an example with a concept such as fairness would guide the developer's interpretation.
An alternative is therefore to present **only the concrete scenario and ask what the developer would do**.
The resulting measure would then not answer "Is this developer fair or unfair?" Instead, it would measure **how closely the developer's decision aligns with the operational profile of fairness elicited by the study**.

Insight: Use concrete scenarios and use examples as leading questions

P_02[Anthropologist]: Formulations beginning with *I prefer* can make the questionnaire appear to measure whether the developer is "ethical" or "unethical." A low score might be interpreted as indicating that the respondent is a bad or unfair developer.
It is also possible that developers routinely implement a behavior without consciously reflecting on its ethical implications. Their questionnaire response could therefore diverge from their actual coding behavior.

P_02[Anthropologist] / ORG_01: If the questionnaire starts directly from concrete scenarios, *I prefer* may be unnecessary and should probably be removed.
Some positively and negatively framed scenarios could also help reduce uniformly high socially desirable responses.

Insight: Avoid morally loaded formulations to reduce positive bias

P_04[DigitalHumanity]: Different statements imply decisions with substantially different costs and degrees of autonomy.
For example, accepting multiple data formats may be a low-cost implementation decision that is almost entirely under the developer's control. Deciding not to collect a piece of user information may instead conflict with requirements or decisions made by other stakeholders.
These two choices therefore cannot automatically be treated as equivalent observations of the developer's ethical profile.

ORG_00 / ORG_01: The current assumption that all statements have equal weight should be reconsidered.

Insight: Statements do not have equivalent weights

P_02[Anthropologist]: If scores are aggregated using a mean, differences in statement weights should be incorporated somehow. Otherwise the resulting profile risks treating qualitatively different decisions as equivalent.
A more qualitative elicitation strategy may therefore be more informative than simply averaging 1-5 scores.

P_04[DigitalHumanity]: Developers should be prompted with qualitative alternatives rather than quantitative 1-5 score.

Insight: Qualitative ranking should be preferred over quantitative scores

---

3. Prompt construction

📝 Participants were explained how the approach proposes to build the prompt. Questions on averaging values were already addressed in the previous discussion. Discussion focused on the information that should be included in the prompt.

P_02[Anthropologist]: The LLM needs explicit information about the context in which the developer profile can be applied. Otherwise, it may use the profile to make decisions based on its own unstated assumptions, including in areas where the developer would not actually have autonomy.
Explicit context would also make the generated output more traceable: it would be easier to identify which constraints and which developer preferences produced a particular choice.

ORG_00: The prompt should therefore first define the **task and its boundaries**, and only afterwards introduce the developer's preferences.
The LLM should:

1. remain within the requirements and constraints of the development task;
2. identify decisions that remain open within those boundaries;
3. apply the developer profile only to those open decisions.

📝 The developer profile cannot be passed to the LLM independently of the development context.

Conceptually, the prompt should contain an explicit **[BOUNDARIES]** component before the developer-profile information.
The profile should not override functional, organizational, legal, or other fixed constraints.

Insight: task boundaries should be explicitly defined

P_05[Developer]: A developer may not have a single profile that applies equally to every task or application. The importance assigned to fairness, sustainability, inclusiveness, or other dimensions may change depending on what is being developed.

ORG_00: This introduces the general issue that an ethical profile depends on context.

📝 The profile may therefore need to be elicited after providing information about the application or project, rather than once and for all.

P_05[Developer]: Application-level context may be more appropriate than individual task-level context. For example, knowing that the developer is working on a healthcare system already establishes relevant characteristics of users, risks, and requirements.

ORG_01: Before administering the ethical statements, the study could therefore elicit or specify the scenario in which development takes place, such as:

- healthcare;
- university or educational setting;
- applications involving young users;
- particular user populations;
- other relevant deployment environments.

The questionnaire would then measure the developer's profile **relative to that application context**.

Insight: the developer profile is context-dependent

📝 The group discussed a methodological trade-off between an overly generic profile and a profile that is excessively fitted to one project.

P_02[Anthropologist]: If the profile is elicited for a specific project, it may no longer represent a general "developer ethical profile." Instead, it represents the developer's preferences for applying ethical orientations in that particular context. Such a profile may not transfer to another project.

P_04[DigitalHumanity]: Context dependence should not necessarily be treated as a flaw. Ethical priorities are inherently contextual. Accepting this complexity may prevent an oversimplified profile that ignores meaningful differences between applications.
The trade-off is therefore between **overfitting to a specific context** and **oversimplifying the developer's ethical profile**.

P_03[SoftwareEng]: An additional research question is what **does not change across contexts**. Even if some aspects of the developer profile vary depending on the application, others may remain stable.
For example, a developer's treatment of fairness-related decisions could remain relatively consistent across different scenarios.
Studying both context-dependent and context-invariant characteristics could therefore be informative, although evaluating this would require multiple development contexts.

Insight: the study should balance context dependence versus stable developer characteristics in the developer profile

📝 Further discussion focused on the actual autonomy held by the developer.

P_01[Lawyer]: The framework should carefully examine how much ethical autonomy a developer really has. Some issues have already been translated into legal requirements or fundamental rights, substantially reducing the space for individual ethical discretion.
The relevant question may therefore be less "What is the developer free to do?" and more **how the developer's ethical profile shapes the interpretation and implementation of requirements in areas where multiple compliant implementations remain possible**.

P_05[Developer]: In practice, developers often have substantial freedom over details that requirements do not specify. For example, how a form is implemented or which alternatives are offered. These choices may be based on experience, conventions, or personal assumptions.
Legal or organizational constraints can reduce this space, but when those constraints do not specify an implementation, developers still frequently make the final decision.

Insight: developers do have autonomy

📝**Developer autonomy is perceived differently across professional perspectives:** The focus group revealed a different perspective between a regulation-centered perspective and a development-practice perspective. The lawyer repeatedly emphasized that legal obligations already prescribe many decisions, thereby leaving developers relatively limited ethical discretion. In contrast, the industry developer stressed that requirements and regulations typically define boundaries without fully specifying implementation, leaving developers substantial room to choose among multiple valid solutions. This suggests that the developer's actual decision space should not be assumed a priori, but explicitly identified as the residual implementation space left open after legal, organizational, functional, and technical constraints have been applied.

Main implications emerging from the focus group

1. **Reframe fairness positively as well as negatively:** include promoting fairness, not only avoiding unfair outcomes.
2. **Separate privacy from fairness:** data collection and privacy deserve independent treatment; do not restrict the discussion to sensitive data alone.
3. **Keep developer agency explicit:** the construct concerns choices made by developers within the implementation space left open by requirements.
4. **Define [BOUNDARIES] before applying the developer profile:** distinguish fixed requirements and constraints from actual developer discretion.
5. **Prevent company-level choices from being attributed to the developer:** the LLM should emulate the developer only within the developer's genuine decision space.
6. **Use concrete and actionable development scenarios in the questionnaire:** abstract ethical statements may not reflect the way developers reason during coding.
7. **Consider placing the example before the statements:** measure the developer's response to a concrete decision rather than asking directly whether they endorse an ethical value.
8. **Interpret scores as adherence to an elicited operational profile, not as whether a developer is ethical/unethical.**
9. **Reconsider equal weighting and simple averaging of questionnaire items:** statements differ in cost, relevance, and degree of developer autonomy.
10. **Include concrete examples in the rationale for each ethical dimension.**
11. **Consider preservation of end-user values as a developer value:** developers may value enabling users' understanding, choice, autonomy, and accessibility.
12. **Elicit application context before the developer profile:** ethical priorities may depend on the application domain, user population, and deployment setting.
13. **Prefer application-level over excessively task-specific contextualization where possible:** this may provide context while reducing overfitting.
14. **Investigate both contextual variation and invariants:** identify which aspects of the developer profile change across applications and which remain stable.
15. **Treat developer autonomy as an empirical issue:** the relevant ethical space consists of implementation decisions that remain open after legal, organizational, functional, and technical constraints have been applied.